Class Meeting Handout #16

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There are many ways to investigate reaction mechanism. Some are obvious: use reaction kinetics or examine the products produced. Others are clever: using competing intramolecular reactions as clocks to study radical reactions. Today we will sample some of these methods.[1]

[1] Read chapter 8.8.

A Tool Box

The strength and enduring power of physical organic chemistry is its contribution to investigating reaction mechanism. Once we know the mechanism and understand how structure affects reactivity we can design reactions that achieve goals thought to be impossible in the dark times before you were born.

Exercise: Tricks of the Trade – 40 minutes

In your reading of the textbook you will have encountered the following methods for elucidating mechanism.

1. **Product Analysis**:[2] The major product is the goal, but the minor products can reveal a lot about the presence of an intermediate or alternate pathways in your mechanism scheme. Propose a reaction where product analysis can confirm your hypothesis for the mechanism.

[2] See chapter 8.8.1

2. **Trapping Intermediates**:[3] If it exists it should be able to be trapped.[4] If a high energy intermediate is proposed in a mechanism then we might be able to observe reaction between that intermediate and a trapping reagent that will produce a predicted alternate product. Propose a reaction where a trapping strategy can confirm your hypothesis for the mechanism.

[3] See chapter 8.8.2 and 8.8.3

[4] So why haven't we trapped Bigfoot?

3. **Stereochemistry**:[5] The classic example is the stereospecificity observed in S_N2 mechanisms. Propose another reaction where stereochemistry can confirm your hypothesis for the mechanism.

[5] See chapter 8.8.6

4. **Radical Clocks**:[6] If we want to investigate the presence of a radical intermediate we could use trapping reagents or we could build into the reactant a group that will react with the radical intermediate in a predictable manner. What if the rate of the alternate route is known to a good degree of accuracy? Now we have a race, and races are decided by clocks. Propose a reaction where this radical clock method can confirm your hypothesis for the mechanism.[7]

[6] See chapter 8.8.8

[7] It is totally OK to use the examples from the textbook for all of these exercises. I want you to read the textbook.

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will…

- come to understand that there are many ways to conduct experiments that can confirm or refute your mechanistic hypothesis.
- be able to design a set of experiments to test a hypothesis for a mechanism.
- be able to criticize the hard work of others as you read literature contributions that investigate the mechanisms of organic chemistry.[8]

[8] It's so much better to be right than to be popular.

A Look Ahead

We have explored a set of tools that can be used investigate reaction mechanisms and to investigate the structure of transition states. These tools have relied on changes in electronic and steric energy or on common-sense reaction kinetics and product analysis. But, there is another way. In the next few class meetings we will explore a different energy regime: vibrational energy.

Electronic changes have a large effect on the energy of a transition state. Changes in steric groups are the same as electronic energy changes. It's all about electrons hating each other. Inside this world of electrons is a subtle but important effect due to mass. Vibrational energy is dependant on the electronic nature of the bond and the mass of the nuclei involved. What if I could investigate a transition state without changing the electronics? What if I only changed the mass? We can do this with isotopes. The next lesson in the course will focus on these isotope effects.